

Method of testing structural integrity on small scale cast-concrete parts

Method of testing

As a method of testing we've chosen collision between static cast-concrete part and conical, flattened tip rod made from high speed steel HSS [Fig 1] mounted on dynamic carriage constrained to linear movement, which will be dropped onto concrete sample.

Magnitude of impact can be adjusted by varying drop height h and-or mass of dynamic carriage m , and will be quantified by kinetic energy E_k expressed by equation (1), where v represents the magnitude of dynamic system's velocity¹.

$$E_k = \frac{1}{2}mv^2 \quad (1)$$

For simplicity we opted to ignore sliding friction between dynamic carriage and linear guides, and air resistance as well since air resistance at velocities we are going to achieve will have negligible effect on final speed.

As far as we are concerned gravitational acceleration g is also constant with value of $9.81 \text{ [ms}^{-2}\text{]}$. Therefor velocity right before impact is dependent only on drop height and can be expressed by equation (2).

$$v = \sqrt{2gh} \quad (2)$$

by substituting (2) into (1) and simplifying we get formula (3) for calculating kinetic energy of dynamic carriage at moment of impact dropped from drop height.

$$E_k = ghm \quad (3)$$

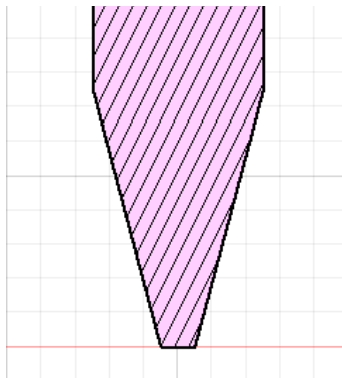


Fig 1: Cross section of HSS tip

¹ since our system is limited to linear movement in one axis we can treat vector quantities such as velocity as single dimensional scalars.

Testing equipment

Requirements

Rigidity

Test rig should be as rigid as possible especially in rotational deflection of dynamic carriage in two axes perpendicular to drop axis. Also there should be minimal translational deflection in plane perpendicular to drop axis. Last but not least there must not be any movement in axial direction between HSS rod and dynamic carriage.

Repeatability

To maintain repeatable results it is necessary to choose proper materials and parts that can handle sudden impact forces without plastic deformation, also regular inspection of the HSS rod is welcomed. Reforming the tip is mandatory if it is severely deformed.

Accuracy

Drop height must be “settable” by some retaining mechanism that will block movement further up. Also there must exist some system for measuring drop height. We highly recommend to do some testing and-or calibration by measuring real time it took to drop dynamic carriage from known height and comparing it to theoretical time t given by formula(4)

$$t = \sqrt{\frac{2h}{g}} \quad (4)$$

Method of time measurement

For measuring time elapsed from release to impact we came up with simple, reliable and precise enough method. Core principle of this method is triggering timer circuit when released from risen position and stopping timer when HSS tip hits the bottom. This is established by connecting dynamic carriage to high signal, bottom and end-stop to measuring device.

t_0 carriage is touching end-stop (circuit is close)

t_s carriage starts to fall (circuit is open)

t_f carriage hits bottom (circuit is closed)

Time of fall is exactly equal to time when circuit is open. Measuring this time can be done ideally via oscilloscope but arduino² will do the job just fine. If previous 2 methods are unavailable for you can still measure it by recording the fall on phone and analyze it frame by frame and count how many frames are between start and finish.

2 Arduino code with manual can be found in attached file stopwatch.ino

Construction

building materials

linear guides – 16 mm cold rolled polished linear rods.

Linear guides fixture - SHF16

Linear bearings – SMA-16GUU

Test results

after testing look for fractures, chipping etc. inspect craters (depth, width).

Results of these tests won't give you tensile strength value nor any useful number. Main purpose of this test is to be quick, simple and to determine whether some concrete mix or casting technique works or not. Also it is good enough to compare concrete samples between each other although the results should be taken with a grain of salt.

Improvements

If you want to move this testing method to another level you can install accelerometer with high enough sample rate on dynamic carriage. With this addition you'll be able to watch the collision develop over time instead of seeing it as an instantaneous event and perceive only aftermath after impact.

Caution

Make sure that dynamic carriage is fully lowered or safely secured before manipulating with sample or doing any other action that would require putting body parts below dynamic carriage. Unscheduled drop on body part can result in serious injury in some cases with permanent consequences depending on mass and drop height.

Safety glasses are necessary due to possibility of fast flying concrete chips in your direction.

Respiratory system protection is also highly advised when handling or testing hazardous materials such as asbestos.

